ARTIFICIAL INTELLIGENCE

CBSE Previous Year Question Paper 2022 • Q.P. Code 104 • Set 4
Answer Key: MCQs, Short Answers & Long Answers

Note: The 2022 paper has three sections. Section A covers Employability Skills, Section B covers subject-specific AI skills, and Section C contains competency-based questions. Answers are provided for all questions for revision.

SECTION A — EMPLOYABILITY SKILLS

Q1–Q4: One-mark questions

Q1. Wage employment is work performed for an employer in return for wages or a salary.

Q2. Food insecurity is a major sustainability concern because people need reliable access to sufficient, safe and nutritious food. Poverty, unequal distribution and environmental damage can make food unavailable or unaffordable.

Q3. Myth. A business idea does not have to be completely unique. A business can succeed by improving an existing product or service, serving customers better or meeting an unmet local need.

Q4. The major purpose of the Sustainable Development Goals (SDGs) is to improve people's lives while protecting the planet, ending poverty and reducing inequality through sustainable development.

Q5–Q6: Two-mark questions

Q5. Write any four common functions of an entrepreneur.

Identifying business opportunities; planning and organising resources; taking calculated risks and making decisions; arranging finance and managing day-to-day operations. Other valid functions include innovation, marketing and coordinating employees.

Q6. Mention four ways individuals can help reduce inequality.

Treat people fairly and respectfully regardless of background; challenge discrimination and stereotypes; support inclusive access to education and opportunities; volunteer or contribute to community initiatives that help disadvantaged people.

SECTION B — SUBJECT-SPECIFIC AI SKILLS

Q7–Q12: One-mark questions

Q7. What is NLP? Natural Language Processing (NLP) is a field of AI that enables computers to process, understand and generate human language.

Q8. Mention two applications of NLP. Examples: language translation and spam/email filtering. Other valid examples include chatbots, sentiment analysis and speech-to-text.

Q9. Name two popular virtual assistants. Siri and Google Assistant. Alexa is another valid example.

Q10. Name the process of dividing a corpus into sentences. Sentence segmentation.

Q11. Define True Positive. A true positive occurs when the model predicts a positive result and the actual result is also positive.

Q12. What is F1 score? F1 score is the harmonic mean of precision and recall: F1 = 2 × (Precision × Recall) / (Precision + Recall).

Q13–Q18: Two-mark questions

Q13. Differentiate between Script-bot and Smart-bot.

A Script-bot follows predefined scripts or rules and handles expected inputs. A Smart-bot uses AI methods such as NLP or machine learning to interpret varied inputs and respond more flexibly.

Q14. What is the purpose of the Evaluation stage of the AI Project Cycle?

Evaluation checks how well the trained model performs on suitable test data. Metrics such as accuracy, precision, recall and F1 score help identify errors and judge whether the model meets the problem's objectives.

Q15. What is tokenization? Count the tokens in the sentence: “I find that the harder I work, the more luck I seem to have.”

Tokenization divides text into smaller units called tokens, such as words or punctuation marks. Counting word tokens only (ignoring punctuation as separate tokens), there are **14 words**: I(1) find(2) that(3) the(4) harder(5) I(6) work(7) the(8) more(9) luck(10) I(11) seem(12) to(13) have(14). If punctuation is counted as separate tokens, the count depends on the tokenizer’s rules.

Q16. Complete the stemming table for “Tries” and “Learning”.

One school-level stemming answer: **Tries** → **affix “-es”; stem “Tri”** and **Learning** → **affix “-ing”; stem “Learn”**. Stemming algorithms can differ; the expected table may depend on the textbook’s specific examples.

Q17. Define Accuracy and give its formula.

Accuracy is the proportion of all predictions that are correct. Formula: Accuracy = (TP + TN) / (TP + TN + FP + FN) × 100%.

Q18. Identify the NLP process in the image and state its purpose.

The image shows **case normalisation / lowercasing**: different capitalisation variants such as “HELLO”, “HeLLo” and “hELLO” are converted to “hello”. Its purpose is to standardise text so that capitalisation differences do not cause the same word to be treated as different tokens.

SECTION C — COMPETENCY-BASED QUESTIONS

Q19. Apply the four steps of Bag of Words (BoW) to the three documents.

Documents:
D1: Sahil likes to play cricket
D2: Sajal likes cricket too
D3: Sajal also likes to play basketball

Step 1 — Text cleaning: Convert text to lowercase and remove punctuation (none is present).
Step 2 — Tokenisation: Split each document into words.
Step 3 — Vocabulary: Use the unique words in first-occurrence order.
Step 4 — Vectorisation: Count each vocabulary word in each document.

Vocabulary	D1	D2	D3
sahil	1	0	0
likes	1	1	1
to	1	0	1
play	1	0	1
cricket	1	1	0
sajal	0	1	1
too	0	1	0
also	0	0	1
basketball	0	0	1

Q20. Explain Term Frequency and Inverse Document Frequency with examples.

Term Frequency (TF): Measures how often a term occurs in a document. A common simple formula is TF = (number of occurrences of the term) / (total number of terms in that document).
Inverse Document Frequency (IDF): Measures how rare a term is across the document collection. A common form is IDF(t) = log(N / df(t)), where N is the number of documents and df(t) is the number of documents containing term t. A word appearing in many documents gets a lower IDF; a rarer word gets a higher IDF. TF-IDF combines both to highlight terms important to a particular document.

Q21. Calculate the F1 score from the traffic-jam confusion matrix.

From the table: TP = 50, FP = 50, FN = 0, TN = 0 (taking Actual = 1 as the positive class).
Precision = TP / (TP + FP) = 50 / (50 + 50) = **0.50**.
Recall = TP / (TP + FN) = 50 / (50 + 0) = **1.00**.
F1 = 2 × (Precision × Recall) / (Precision + Recall) = 2 × (0.5 × 1) / (0.5 + 1) = **0.6667** ≈ **66.67%**.

Important notes

• Q15 token count: the answer above counts words only (14). Some tokenizers count punctuation separately, which changes the total.
• Q16 stemming can vary by algorithm and by the textbook's convention. The suggested stems illustrate a simple suffix-removal approach.
• Q21 values are read from the matrix on page 7: TP=50, FP=50, FN=0 and TN=0.